FalseGreen

VERIFICATION · SIGNED EVIDENCE

# FalseGreen Verification Report

**ACCEPTED**

All required criteria passed for the exact source tree built and verified by FalseGreen.

This does not prove universal correctness. Assurance is limited to the explicit formal properties, assumptions, models, summaries, and scope recorded in this report.

## THE WORK

| | |
|---|---|
| Job | `0744390a-a793-4c99-9df7-0c948d247991` |
| Language | rust |
| Toolchain | rustc 1.99.0 (rustc) (1.99.0) |
| Verifier environment | rust - rustc 1.99.0 (rustc) (1.99.0) |

## THE FROZEN BOUNDARY

| | |
|---|---|
| Source tree | `93e71980278736e3f889810d7527966f1684833b5b66649bacca8d76a6ff3a9a` |
| Contract SHA-256 | `f6aa9bbaf7f192aff07ebe0fd0fdeae49ec70abbdf24dfdb05e2a37492002d82` |
| Criteria SHA-256 | `8732560e4c69b482eee5227ae126766b4f2ee7558fd4f77d90b5564776a83af4` |
| Criteria | 3 / 3 PASSED |

## WHAT FALSEGREEN DID

| | |
|---|---|
| Attempts | 1 |
| Duration | 1m 28s |
| Report schema | v12 |
| Report record | 1 (the verdict mint) |
| Signed | 2026-10-25 22:05:38 UTC |

## Contract (frozen)

Wall budget 2700s

| Criterion | Title | Required | Result | Duration |
|---|---|---|---|---|
| `build` | the workspace compiles (release) | yes | **PASS** | 20.4s |
| `tests` | the full test suite passes offline | yes | **PASS** | 11.5s |
| `formatting` | the tree is rustfmt-clean | no | **PASS** | 0.1s |

Invariants chain linking is exact successor arithmetic and never wraps at the u64 ceiling — `ladder.l1-kani, ladder.l2-creusot`

## Plan (frozen)

| | |
|---|---|
| Objective | Produce the 0.14.0 marketing verification of the rust-demo ledger workspace through the proof ladder: the mechanical floor (compile, full test suite, format ting) as contract criteria, and exactly one claim about the program — 'chain linking is exact successor arithmetic and never wraps at the u64 ceiling' — carried by two formal encodings the machines discharge: Kani L1 (bounded model check over the SHIPPED arithmetic in ledger-rules, full u64 domain) and Creusot L2 (unbounded Why3 proof of the guarded spec inside crates/le dger-spec). The 0.11.0-era ten claims that 'proved' themselves with a mechanical check are deliberately NOT carried: a passing test proves the test passed, so a claim without a formal encoding leaves the contract (user ruling 2026-10-25). The signed report is the artifact (the site + marketing surface needs a current accepted report on the shipped product). |
| Plan digest | `e8de5eab37d325cff9de4e07681b4610c010bd57a9ed86fe637e1b538a24f14f` |
| Work items | 4 |

Work items, in dependency order:

| Work item | Outcome | Depends on | Omission risk |
|---|---|---|---|
| `criteria-floor` | The mechanical floor runs as criteria on every attempt: the workspace compiles in release, the full test suite passes, the tree is rustfmt-clean (opt ional). | - | a criterion that matches nothing would pass vacuously — each argv is the whole suite or a build, not a test-name filter |
| `ladder.l1-kani` | Kani L1 decides the claim over the SHIPPED arithmetic (proofs/links_ka ni.rs pulls crates/ledger-rules/src/lib.r s in by path, so the bytes the binary runs are the bytes proved): the successor predicate never wraps at the u64 ceiling, and on the guarded domain it is exactly seq == prev + 1. | criteria-floor | a harness that copies the code instead of including it would prove a copy, not the shipped arithmetic; a harness that explodes in std machinery would come back UNAVAILABLE and hold the Job at insufficient_evidence |
| `ladder.l2-creusot` | Creusot L2 proves the guarded spec (#[requires(prev < u64::MAX)] + #[en sures(result == (seq == prev + 1u64))] inside crates/ledger-spec) for ALL inputs — unbounded, so it covers what the bounded check cannot. | criteria-floor | an empty or non-matching target list yields zero goals — the rung must be recorded, never silently passed; the workspace-root sr c/ scan does not see a member crate's spec (0.14.0 finding #33), so the claim TEXT may be missing from the row even though the proof closed |
| `report-artifact` | The Job terminates `accepted` with both rung rows and all five security classes satisfied; the signed JSON + the human rendering become the marketing artifact (saved in the repo, locally verified keyless). | ladder.l1-kani, ladder.l2-creusot | a security-class shortfall (secret/SCA/SC) would verdict the Job insufficient_evidence or failed regardless of the proofs |

Acceptance conditions, and the evidence that decides each:

| Scope | Condition | Evidence |
| --- | --- | --- |
| `criteria-floor` | the release build of the whole workspace succeeds inside the job container | build |
| `criteria-floor` | the full workspace test suite passes in the job container (the shipped ledger behaviour, unfiltered) | behavioral-test |
| `ladder.l1-kani` | the kani row in the signed report decides `proven` (every harness check discharged) with its extracted clauses and encoding_sha256 recorded | proof |
| `ladder.l2-creusot` | the creusot row decides `proven` with every generated verification condition closed (goals 1/1) and its derived trust set recorded | proof |
| `report-artifact` | the terminal verdict is `accepted`, the report carries both invariant rows with decisions, and `falsegreen_verify_report` verifies the saved JSON locally (Ed25519, keyless) | artifact |
| `overall` | one Job, one entitlement, terminal `accepted`: the mechanical criteria green, the Kani rung proven (bounded), the Creusot rung proven (unbounded), all five mandatory security classes satisfied, and the saved signed report verifies locally. | proof |

## Evidence manifest

### ATTEMPT 1 - PASSED - 2026-10-25 22:05:38 UTC - WALL 1M 28S

source sha256: 93e71980278736e3f889810d7527966f1684833b5b66649bacca8d76a6ff3a9a

env sha256: 39692e8bb1a7dc5e464d1fb48e19afdc2d7a8ace836553f780fc660e3897a4ab

## Security Findings

The bar these findings are judged against (the frozen policy): **fail on:** critical · **unknown severity:** insufficient_evidence.

*Nothing to act on — 180 recorded facts (are inventory evidence in the signed JSON report).*

The full SBOM and VEX are in the signed JSON report.

JSON report: https://falsegreen.com/report/0744390a-a793-4c99-9df7-0c948d247991/json

## Invariant Checks

Each entry is ONE frozen claim, the PROVER that worked on it, and the formal proposition that prover was asked to discharge — read out of the frozen encoding, so the machine's text sits next to the author's sentence. The full rows (any counterexample trace, the admitted assumptions) are in the signed JSON report.

### ATTEMPT #1

#### chain linking is exact successor arithmetic and never wraps at the u64 ceiling

**PROVEN**

"record N links to record N-1 iff prev_hash equals record N-1's hash AND seq is the exact u64 successor of record N-1's seq; the successor is never taken modulo the u64 ceiling — no seq may link past u64::MAX (the chain cannot wrap) — and on the guarded domain (prev below u64::MAX) the predicate is exactly seq == prev + 1"

user-provided description — not machine-checked

Formal claim:

```
kani::assume(seq < u64::MAX)
kani::assert(!wrapped)
kani::assume(prev < u64::MAX)
kani::assert(got == (seq == prev.wrapping_add(1)))
```

Harness: `successor_never_wraps_at_the_u64_ceiling`

`encoding sha256: e87bb002ddad5cf216c2e7df97470f826a80205820eff0e66ffd16bcf5c19d24`

decided by: kani Rust Verifier 0.68.0 (standalone) (bounded)

## chain linking is exact successor arithmetic and never wraps at the u64 ceiling

**PROVEN 1 / 1**

"record N links to record N-1 iff prev_hash equals record N-1's hash AND seq is the exact u64 successor of record N-1's seq; the successor is never taken modulo the u64 ceiling — no seq may link past u64::MAX (the chain cannot wrap) — and on the guarded domain (prev below u64::MAX) the predicate is exactly seq == prev + 1"

user-provided description — not machine-checked

Target: `verif/ledger_spec_rlib/links`

`encoding sha256: e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855`

decided by: creusot cargo 1.99.0 (5f94df478 2026-08-27) (unbounded) on alt-ergo, z3

## Security

| | |
|---|---|
| Policy | w9-marketing |
| Policy digest | `5600a5936458417049781c8112ea8caeaff5f23b86815555880e7b0c80780213` |
| Result | SATISFIED |

| Evidence class | Required | Status | Reasons |
|---|---|---|---|
| Dependency/SCA | yes | **PASS** | |
| Static analysis | yes | **PASS** | |
| Secrets | yes | **PASS** | |
| Supply chain | yes | **PASS** | 92 components inventoried (evidence, in the signed JSON report) |
| License policy | yes | **NOT ASSESSED** | no licence policy frozen — the licence facts are inventory, nothing was asserted · 88 licence facts inventoried (evidence, in the signed JSON report) |

**How to read this table.** **PASS** — the evidence was collected and nothing in it violated the frozen policy. **FAIL** — the evidence showed a violation the policy forbids; that alone fails the Job. **INSUFFICIENT EVIDENCE** — the evidence could not be collected, or could not be decided for a reason outside the code (a stale snapshot, an unavailable tool, output the parser could not read). It is never a pass, and it holds the Job at *insufficient_evidence* rather than accepting it. **NOT ASSESSED** — no policy is frozen for this class, so nothing was asserted about it; the facts are recorded as inventory and the class does not hold the Job. The findings table lists only what the frozen policy JUDGED — a graded severity, supply-chain drift, or a licence breaking the frozen policy. Everything else the tools recorded (components, licence facts) is inventory: not a finding, counted on the class row above, kept in the signed JSON report. A finding is **OPEN** unless it is resolved, not affected, or covered by a recorded exception.

Attempt security history (the full per-attempt sections are in the signed JSON report):

| Attempt | Result | Open findings | Demonstrated violations |
|---|---|---|---|
| #1 | **SATISFIED** | 0 | — |

**Witnesses (tool provenance)**

```
osv-scanner 2.7.0 (Dependency/SCA) — invocation sha256:
835b11848c29600837dde9101735aa2b1fd877a45db0bbabb127ab134fe1b0c1 —
39692e8bb1a7dc5e464d1fb48e19afdc2d7a8ace836553f780fc660e3897a4ab
```

semgrep 1.181.0 (Static analysis) — invocation sha256: a47092b39c1cba80033911904dd2c9e75edbfd6203514480334a063f1296d10b — 39692e8bb1a7dc5e464d1fb48e19afdc2d7a8ace836553f780fc660e3897a4ab

gitleaks 8.30.1 (Secrets) — invocation sha256: 897c6b806ddd3f6bfc1699edeb40c3a775a57681925a6e7f867281c456e297d3 — 39692e8bb1a7dc5e464d1fb48e19afdc2d7a8ace836553f780fc660e3897a4ab

syft 1.54.1 (Supply chain) — invocation sha256: b2d1b394eee5e3e6fe2843bfe2b2a0b83babebcc4b5d14a23b1a9e71ce885312 — 39692e8bb1a7dc5e464d1fb48e19afdc2d7a8ace836553f780fc660e3897a4ab

dependency-manifest 1 (License policy) — invocation sha256: 63ab5d8d0fcd155c6dd7130156c7b8444ca286819623df2e4eb2e27cc387acf7 — 39692e8bb1a7dc5e464d1fb48e19afdc2d7a8ace836553f780fc660e3897a4ab

## Tools & provers used

| Tool | Version | What it did in this run |
|---|---|---|
| **osv-scanner** | `2.7.0` | Dependency/SCA evidence witness |
| **semgrep** | `1.181.0` | Static analysis evidence witness |
| **gitleaks** | `8.30.1` | Secrets evidence witness |
| **syft** | `1.54.1` | Supply chain evidence witness |
| **dependency-manifest** | `1` | License policy evidence witness |
| **kani** | `Kani Rust Verifier 0.68.0 (standalone)` | invariant proofs (1 row) |
| **creusot** | `cargo 1.99.0 (5f94df47 8 2026-08-27)` | invariant proofs (1 row) |
| **alt-ergo, z3** | `not recorded (image-pi nned)` | authorized provers (frozen contract) |

## Exports

Full content is embedded in the signed JSON report (report schema v12).

`spdx-3.0 sha256:3ca7d99f4c31bb34a250ab1064c107a3b4694dd4866eee15ad5613992d9b7c07 embedded in the JSON report (66947 bytes)`

`cyclonedx-1.7 sha256:6c6fb6eeca87b6b9045d12eaa608cdf94ad048e8c3f4d63baf9b6fe6ae67e587 embedded in the JSON report (32209 bytes)`

`cyclonedx-1.7-vex sha256:c637993a00cbd83087869ed9a40c56a1800a8f18b7820db5312a9d77a6a30eb4 embedded in the JSON report (632 bytes)`

## Signature

This report is cryptographically signed: Ed25519 over the canonical report content. The public key is embedded in the report — verification is offline and registry-free (`falsegreen verify-report`).

```
key id: fg-report-9fc1a2c5-134c-4ca1-a0f2-56b2d1bcde67-20261018-1
ed25519 public key: ie4eQdY1lt7O5aCCQzO+/192Yzc4pLQsGyVNqSKtlmA=
ed25519 signature: CfILwToLf1/whW/0tqYpQdX60tlKJVgYoBaP32PSeZFS3J5hyc3o7XvUBl75IIOTeCjZNkvFozHB2rpGYUdeBg==
digest: 8344852cfa4bc11968d4a003f48cf27b53ca5174d0d966db22da47b7edfea4a8
```

This is a presentation of the signed canonical evidence (the JSON artifact). It is not an audit certificate. - FalseGreen v0.14.0